

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location CoreSite SV7, CA -- station KSJC, America/Los_Angeles
Committed pair OSM 1160688107 -> 54173383
CoreSite SV7 / CoreSite SV3
Facade gap 63.6 m between the two halls
Weather record 43,747 real hourly records from KSJC
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3340 C at 315 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-10-22 (chatter)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 2. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 3 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	17.451	18.0	17.220	0.993
01	mechanical	yes	switch budget	17.501	18.0	17.220	0.956
02	mechanical	yes	switch budget	17.943	18.0	18.330	0.745
03	mechanical	yes	switch budget	17.776	18.0	17.780	0.765
04	mechanical	no	dry-bulb	18.280	18.0	18.330	0.931
05	mechanical	no	dry-bulb	18.203	18.0	17.223	0.792
06	mechanical	yes	-	16.998	18.0	14.764	0.799
07	mechanical	no	dry-bulb	18.062	18.0	14.613	1.117
08	mechanical	no	dry-bulb	18.622	18.0	14.613	1.337
09	mechanical	no	dry-bulb	18.617	18.0	14.613	1.883
10	mechanical	no	dry-bulb	19.732	18.0	15.733	1.978
11	mechanical	no	dry-bulb	20.400	18.0	17.506	1.692

12	mechanical	no	dry-bulb	20.437	18.0	18.103	1.713
13	mechanical	no	dry-bulb	20.305	18.0	18.409	1.579
14	mechanical	no	dry-bulb	20.735	18.0	19.177	1.525
15	mechanical	no	dry-bulb	19.515	18.0	18.061	1.401
16	mechanical	no	dry-bulb	19.026	18.0	18.611	1.169
17	mechanical	no	dry-bulb	18.183	18.0	16.864	1.330
18	FREE	yes	-	16.421	18.0	16.131	1.252
19	FREE	yes	-	16.248	18.0	15.101	1.522
20	FREE	yes	-	15.258	18.0	14.727	1.580
21	FREE	yes	-	15.306	18.0	14.541	1.571
22	FREE	yes	-	14.780	18.0	14.453	1.190
23	FREE	yes	-	14.553	18.0	14.569	0.903

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.451 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.501 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.943 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.776 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.281 C against a 18.0 C limit, so it fails by 0.281 C. It would take a limit 0.281 C higher, or a bound 0.281 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.203 C against a 18.0 C limit, so it fails by 0.203 C. It would take a limit 0.203 C higher, or a bound 0.203 C tighter, to change this hour.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.999 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the

switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.062 C against a 18.0 C limit, so it fails by 0.062 C. It would take a limit 0.062 C higher, or a bound 0.062 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.622 C against a 18.0 C limit, so it fails by 0.622 C. It would take a limit 0.622 C higher, or a bound 0.622 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.617 C against a 18.0 C limit, so it fails by 0.617 C. It would take a limit 0.617 C higher, or a bound 0.617 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.732 C against a 18.0 C limit, so it fails by 1.732 C. It would take a limit 1.732 C higher, or a bound 1.732 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.400 C against a 18.0 C limit, so it fails by 2.400 C. It would take a limit 2.400 C higher, or a bound 2.400 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.437 C against a 18.0 C limit, so it fails by 2.437 C. It would take a limit 2.437 C higher, or a bound 2.437 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.305 C against a 18.0 C limit, so it fails by 2.305 C. It would take a limit 2.305 C higher, or a bound 2.305 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.735 C against a 18.0 C limit, so it fails by 2.735 C. It would take a limit 2.735 C higher, or a bound 2.735 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.515 C against a 18.0 C limit, so it fails by 1.515 C. It would take a limit 1.515 C higher, or a bound 1.515 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.026 C against a 18.0 C limit, so it fails by 1.026 C. It would take a limit 1.026 C higher, or a bound 1.026 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.183 C against a 18.0 C limit, so it fails by 0.183 C. It would take a limit 0.183 C higher, or a bound 0.183 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.421 C, which is 1.579 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.252 C, and the margin is measured, not chosen: 0.943 C of group-conditional forecast error for this hour of day, plus 0.3096 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.248 C, which is 1.752 C

under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.522 C, and the margin is measured, not chosen: 1.163 C of group-conditional forecast error for this hour of day, plus 0.3586 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.258 C, which is 2.742 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.580 C, and the margin is measured, not chosen: 1.323 C of group-conditional forecast error for this hour of day, plus 0.2576 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.306 C, which is 2.694 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.571 C, and the margin is measured, not chosen: 1.262 C of group-conditional forecast error for this hour of day, plus 0.3096 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.780 C, which is 3.220 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.190 C, and the margin is measured, not chosen: 0.860 C of group-conditional forecast error for this hour of day, plus 0.3294 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.553 C, which is 3.447 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.903 C, and the margin is measured, not chosen: 0.608 C of group-conditional forecast error for this hour of day, plus 0.2955 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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